Isospectral Arithmetic Fiber Retractions Across the $1/h < \Re s \leq 1$ Similarity Gap

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Abstract

For every integer $h \geq 2$, we compare two idempotent arithmetic retractions onto the h-free integers: one caps each prime exponent at $h-1$, while the other reduces it modulo h. Weighting the induced basis maps on $\ell^2(\mathbb{N})$ by $n^{-s/2}$ produces compact operators $S_{h,s}$ and $M_{h,s}$ on their respective domains $\Re s > 0$ and $\Re s > 1/h$. Their simple nonzero eigenvalues coincide, and so do all common legal power traces and integer-order regularized determinants. Nevertheless, $S_{h,s}$ is boundedly similar to a compact normal operator exactly when $\Re s > 1$, whereas $M_{h,s}$ is so similar throughout its bounded domain. Thus the full band $1/h < \Re s \leq 1$ is isospectral but not similar.

The discrepancy is quantified in three ways. Saturated Riesz projections have an exact primorial optimizer and subcritical, critical, and supercritical maximal-order regimes. Both singular-value sequences satisfy Weyl laws with explicit Euler-product constants, meeting at the exact crossover $C_{h,1} = D_{h,1} = 1$. Finally, the self-commutators have the sharp wall $\Re(s)q = 1$, distinct from the operator Schatten wall $\Re(s)q = 2$; the case $h = 2$ requires a separate endpoint witness and admits an exact Hilbert–Schmidt Euler identity. Independent finite and analytic recomputations are reported only as implementation checks: all infinite claims are established by proof.

Contents

1 Introduction

Compact operators can share every nonzero eigenvalue and still have incompatible geometry. The simplest obstruction is already visible in a rank-one block: the eigenvalue records the action on its eigenline, whereas the angle between the range and kernel controls the norm of its Riesz projection. The point of this paper is that two elementary arithmetic retractions organize that obstruction into an exact family of phase transitions.

Fix $h \geq 2$. Let $\mathcal{F}_h$ be the positive integers whose prime exponents are strictly smaller than h, and define

$$\tau_h(n) = \prod_p p^{\min\{v_p(n), h-1\}}, \qquad \omega_h(n) = \prod_p p^{v_p(n) \bmod h}. \tag{1}$$

Both maps retract $\mathbb{N}$ onto $\mathcal{F}_h$. On the finitely supported sequences set

$$S_{h,s} e_n = n^{-s/2} e_{\tau_h(n)}, \qquad M_{h,s} e_n = n^{-s/2} e_{\omega_h(n)}, \qquad \sigma = \Re s. \tag{2}$$

Here and below $n^{-s/2} = \exp(-(s/2)\log n)$ with the real logarithm. The formulas in equation (2) are algebraic until boundedness is proved. The symbols $S_{h,s}$ and $M_{h,s}$ denote operators on $\ell^2(\mathbb{N})$ only on their respective bounded domains; every spectral, ideal, trace, determinant, similarity, and commutator assertion is confined to those domains. For $0 < q < 1$, $\mathcal{S}_q$ denotes the usual quasi-Schatten ideal.

The common fixed points in $\mathcal{F}_h$ force common eigenlines. The fibers above those points are very different: saturation can increase only primes already at exponent $h-1$, whereas reduction modulo h permits an arbitrary hth-power multiplier. That distinction changes fiber mass and block angle without changing the eigenvalue on the fixed point.

Theorem 1.1 (Paired arithmetic retraction theorem)**.** *Let $h \geq 2$, $s \in \mathbb{C}$, and $\sigma = \Re s$.*

(i) *$S_{h,s}$ extends boundedly exactly for $\sigma > 0$, and $M_{h,s}$ extends boundedly exactly for $\sigma > 1/h$. Each extension is compact throughout its bounded domain.*

(ii) *For $k \geq 1$ and $0 < q < \infty$,*

$$S_{h,s}^k \in \mathcal{S}_q \iff k\sigma q > 2,$$
$$M_{h,s}^k \in \mathcal{S}_q \iff \sigma > 1/h \text{ and } k\sigma q > 2.$$

(iii) *On each bounded domain the simple nonzero eigenvalues are $\lambda_m = m^{-s/2}$, $m \in \mathcal{F}_h$. On the common domain $\sigma > 1/h$, if $k\sigma > 2$, then*

$$\operatorname{Tr}(S_{h,s}^k) = \operatorname{Tr}(M_{h,s}^k) = \frac{\zeta(ks/2)}{\zeta(hks/2)}. \tag{3}$$

If $r \geq 1$ is an integer, $\sigma > 1/h$, and $r\sigma > 2$, then

$$\det_r(I - zS_{h,s}) = \det_r(I - zM_{h,s}) \tag{4}$$

as entire functions of z. For $r = 1$, this is the ordinary Fredholm determinant and requires $\sigma > 2$.

(iv) *The exact normal-similarity domains are*

$$S_{h,s} \sim_{\mathrm{bd}} \text{ a compact normal operator} \iff \sigma > 1,$$
$$M_{h,s} \sim_{\mathrm{bd}} \text{ a compact normal operator} \iff \sigma > 1/h.$$

Consequently $1/h < \sigma \le 1$ is an isospectral band in which $M_{h,s}$ is similar to normal and $S_{h,s}$ is not, while the two operators have equal order-r regularized determinants for every integer r that is legal for both. The order $r = 1$ is never legal inside this band, since the Fredholm determinant requires $\sigma > 2$.

(v) *Let $P_k = p_1 \cdots p_k$, with $P_0 = 1$, and take the largest $k = k(x)$ for which $P_k^{h-1} \le x$. The largest saturated Riesz norm among $m \in \mathcal{F}_h$, $m \le x$, is attained at P_k^{h-1}. As $x \to \infty$ this maximum tends to $\sqrt{\zeta(\sigma)}$ if $\sigma > 1$; it is asymptotic to $\sqrt{\mathrm{e}^{\gamma} \log\log x}$ if $\sigma = 1$; and if $0 < \sigma < 1$, its logarithm is asymptotic to*

$$\frac{(h-1)^{\sigma-1}(\log x)^{1-\sigma}}{2(1-\sigma)\log\log x}. \tag{5}$$

(vi) *The singular values, counted with multiplicity in decreasing order, satisfy*

$$s_n(S_{h,s}) \sim \left(\frac{C_{h,\sigma}}{n}\right)^{\sigma/2} \qquad (\sigma > 0),$$
$$s_n(M_{h,s}) \sim \left(\frac{D_{h,\sigma}}{n}\right)^{\sigma/2} \qquad (\sigma > 1/h),$$
$$|\lambda_n| \sim \left(\frac{1/\zeta(h)}{n}\right)^{\sigma/2},$$

where

$$C_{h,\sigma} = \prod_p (1-p^{-1}) \left[\sum_{e=0}^{h-2} p^{-e} + p^{-(h-1)}(1-p^{-\sigma})^{-1/\sigma}\right], \tag{6}$$

$$D_{h,\sigma} = \frac{\zeta(h\sigma)^{1/\sigma}}{\zeta(h)}. \tag{7}$$

The constants meet at $C_{h,\sigma} = D_{h,\sigma} = 1$ when $\sigma = 1$. No ordering between them is asserted away from that point.

(vii) *For $0 < q < \infty$,*

$$[S_{h,s}^*, S_{h,s}] \in \mathcal{S}_q \iff \sigma q > 1,$$
$$[M_{h,s}^*, M_{h,s}] \in \mathcal{S}_q \iff \sigma > 1/h \text{ and } \sigma q > 1.$$

The theorem separates three ledgers. The cyclic ledger—eigenvalues, legal traces, and legal determinants—is common. The metric ledger—block singular values and their Weyl constants—is not. The angular ledger—Riesz projection norms and self-commutators—produces a further pair of sharp walls. In particular, equality of the regularized determinants in equation (4) is a negative control: it demonstrates how much geometry a complete nonzero-eigenvalue product can fail to see.

Relation to prior work. Weighted composition operators on weighted sequence spaces have general boundedness, compactness, closed-range, and essential-norm theories [LK15, AM23]. Carlson's discrete setting gives spectral and commutant context for weighted composition operators on L^2 spaces [Car90]. Our basis maps are naturally viewed on the adjoint side of that general framework; this convention and the generic fiber-norm mechanism are not contributions. Power-free parts also have a classical arithmetic life independent of the operators considered here [dWvdW99]. The residual result is the paired all-h classification and its exact arithmetic distortion laws, not a generic weighted-composition theorem and not a claim that the formulas detect the additive semantics of rational primes.

We use standard trace-ideal and regularized-determinant theory from [Sim05], classical prime asymptotics from [MV06], and Wiener–Ikehara in the form discussed in [Kor04]. The hypotheses needed here are checked explicitly.

Proof organization. Section 2 derives both fibers and their rank-one blocks. Section 3 proves existence, compactness, ideal membership, and the common cyclic ledger. Section 4 identifies the precise similarity gap; sections 5 to 7 quantify its maximal, asymptotic, and commutator manifestations. Independent recomputation is summarized only after the proofs in section 8. It checks the implementation but is not used to prove an endpoint or asymptotic statement.

2 Arithmetic fibers and rank-one blocks

Write

$$\mathcal{F}_h = \{m \in \mathbb{N} : v_p(m) < h \text{ for every prime } p\}, \qquad J_h(m) = \{p : v_p(m) = h-1\}.$$

The set $J_h(m)$ records the prime exponents at which saturation has lost information. It need not equal the set of all prime divisors of m.

Lemma 2.1 (Exact fibers)**.** *For every $m \in \mathcal{F}_h$,*

$$\tau_h^{-1}(m) = \left\{ m \prod_{p \in J_h(m)} p^{r_p} : r_p \in \mathbb{N} \cup \{0\} \right\}, \tag{8}$$

$$\omega_h^{-1}(m) = \{m a^h : a \in \mathbb{N}\}. \tag{9}$$

Both maps fix m.

Proof. If $v_p(m) \le h-2$, the equality $\min\{v_p(n), h-1\} = v_p(m)$ forces $v_p(n) = v_p(m)$. If $v_p(m) = h-1$, it forces only $v_p(n) \ge h-1$, leaving a nonnegative extra exponent at that same prime. No new prime can occur, which proves equation (8). On the other hand, $v_p(n) \equiv v_p(m) \pmod h$, with $0 \le v_p(m) < h$, is equivalent to $v_p(n) = v_p(m) + h v_p(a)$ for a unique positive integer a. This proves equation (9). Taking every extra exponent to be zero shows that both maps fix m. $\square$

For $f = \tau_h$ or ω_h, let

$$\mathcal{H}_m^f = \overline{\operatorname{span}}\{e_n : f(n) = m\}.$$

The fibers partition $\mathbb{N}$, so

$$\ell^2(\mathbb{N}) = \bigoplus_{m \in \mathcal{F}_h} \mathcal{H}_m^f. \tag{10}$$

Lemma 2.2 (Block ledger)**.** *Let T denote either algebraic map in equation* (2)*, and write $T_m = T|_{\mathcal{H}_m^f}$. Whenever the coefficient sequence of this block is square summable, T_m is rank one, its range is $\mathbb{C}e_m$, and its unique nonzero singular value $\rho_T(m)$ satisfies*

$$\rho_S(m)^2 = m^{-\sigma} \prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1}, \qquad \sigma > 0, \tag{11}$$

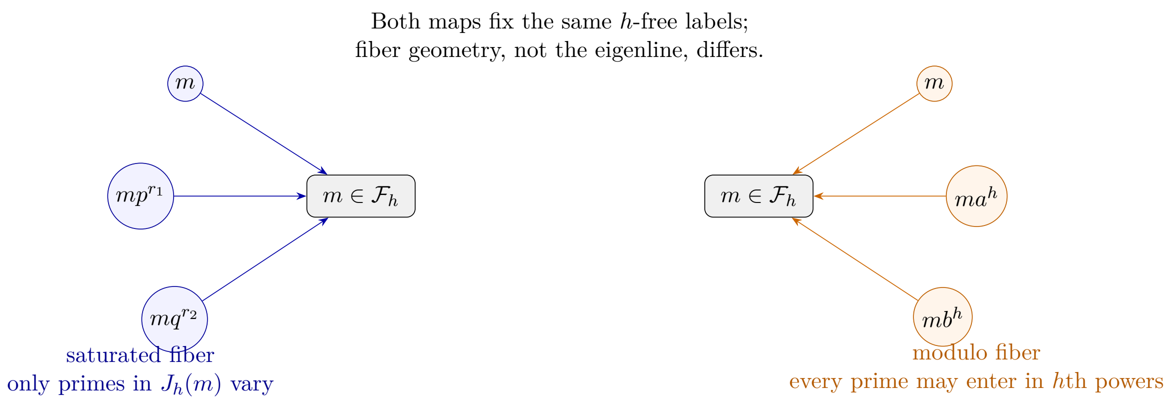

Figure 1: The common fixed label produces the same eigenline. The saturated fiber varies only primes in $J_h(m)$, whereas the modulo fiber permits an arbitrary hth-power multiplier.

$$\rho_M(m)^2 = m^{-\sigma}\zeta(h\sigma), \qquad \sigma > 1/h. \tag{12}$$

Moreover, with $\lambda_m = m^{-s/2}$*,*

$$T_m e_m = \lambda_m e_m, \qquad T_m^k = \lambda_m^{k-1} T_m \quad (k \geq 1). \tag{13}$$

Proof. For a finitely supported $x = \sum_n x_n e_n$ in the block,

$$T_m x = \left(\sum_{f(n)=m} x_n n^{-s/2} \right) e_m.$$

Thus the squared norm of the coefficient functional is $\sum_{f(n)=m} n^{-\sigma}$, and this is the only nonzero squared singular value. Applying lemma 2.1,

$$\sum_{\tau_h(n)=m} n^{-\sigma} = m^{-\sigma} \prod_{p \in J_h(m)} \sum_{r \geq 0} p^{-r\sigma},$$
$$\sum_{\omega_h(n)=m} n^{-\sigma} = m^{-\sigma} \sum_{a \geq 1} a^{-h\sigma},$$

which gives equations (11) and (12) with their exact convergence conditions. Because m lies in its own fiber, $T_m e_m = \lambda_m e_m$. A rank-one map with range $\mathbb{C}e_m$ then obeys $T_m^2 = \lambda_m T_m$, and induction proves equation (13). □

Two type distinctions will be used repeatedly. The eigenvalue λ_m depends on the full complex parameter s; the block singular value depends only on σ. Also, the block index m is an h-free integer, not a prime. Confusing either pair erases the phenomenon we are trying to measure.

3 Existence, ideal membership, and the common cyclic ledger

The block decomposition converts operator questions into positive sums. This also makes every strict endpoint visible without analytic continuation or finite truncation.

Proposition 3.1 (Exact boundedness and compactness domains)**.** *$S_{h,s}$ is bounded exactly when $\sigma > 0$, and $M_{h,s}$ is bounded exactly when $\sigma > 1/h$. Each bounded operator is compact.*

Proof. If $\sigma \le 0$, the saturated block $m = p^{h-1}$ contains p^{h-1+r} for every $r \ge 0$, and its coefficient square mass diverges. Thus $S_{h,s}$ cannot be bounded. Suppose $\sigma > 0$. A nonsaturated local contribution to equation (11) is $p^{-e\sigma} < 1$. A saturated contribution is

$$b_p = p^{-(h-1)\sigma}(1 - p^{-\sigma})^{-1}.$$

Since $b_p \to 0$, only finitely many b_p exceed one; their product is a uniform bound for all squared block norms. Hence $S_{h,s}$ is bounded.

As $m \to \infty$ through $\mathcal{F}_h$, at least one prime divisor of m tends to infinity because every exponent is at most $h-1$. Its local contribution tends to zero, while all other contributions are bounded by the fixed product of the finitely many exceptional factors. Therefore $\rho_S(m) \to 0$. An orthogonal direct sum of rank-one blocks is compact exactly when its block norms tend to zero.

For $M_{h,s}$, the block at $m = 1$ has squared mass $\zeta(h\sigma)$, which is finite exactly for $\sigma > 1/h$. On this domain $\rho_M(m) = \zeta(h\sigma)^{1/2} m^{-\sigma/2} \to 0$, giving both boundedness and compactness. $\square$

Proposition 3.2 (Powers and Schatten walls)**.** *For $k \ge 1$ and $0 < q < \infty$,*

$$S^k_{h,s} \in \mathcal{S}_q \iff k\sigma q > 2,$$
$$M^k_{h,s} \in \mathcal{S}_q \iff \sigma > 1/h \text{ and } k\sigma q > 2.$$

Every equality endpoint fails.

Proof. For $k = 1$, orthogonality and equation (11) give

$$\sum_{m\in\mathcal{F}_h} \rho_S(m)^q = \prod_p \left[1 + \sum_{e=1}^{h-2} p^{-e\sigma q/2} + p^{-(h-1)\sigma q/2}(1 - p^{-\sigma})^{-q/2}\right]. \tag{14}$$

When $h \ge 3$, the first nonconstant term is $p^{-\sigma q/2}$; when $h = 2$, the saturated term is asymptotic to the same quantity. The positive Euler product converges exactly for $\sigma q/2 > 1$.

For M,

$$\sum_{m\in\mathcal{F}_h} \rho_M(m)^q = \zeta(h\sigma)^{q/2} \sum_{m\in\mathcal{F}_h} m^{-\sigma q/2} = \zeta(h\sigma)^{q/2} \frac{\zeta(\sigma q/2)}{\zeta(h\sigma q/2)}. \tag{15}$$

The first factor imposes the boundedness condition $\sigma > 1/h$; the positive h-free sum converges exactly for $\sigma q/2 > 1$.

By equation (13), the unique singular value of T^k_m is $|\lambda_m|^{k-1}\rho_T(m)$. Repeating the two products replaces the leading exponent $\sigma q/2$ by $k\sigma q/2$ while leaving the modulo fiber factor $\zeta(h\sigma)^{q/2}$ unchanged. This proves the stated conditions. At equality, the leading positive prime sum is harmonic over the primes, so cancellation cannot rescue the endpoint. $\square$

Proposition 3.3 (Common spectrum and traces)**.** *On either bounded domain, the nonzero spectrum is the simple set $\{m^{-s/2} : m \in \mathcal{F}_h\}$. On the common bounded domain, if $k\sigma > 2$, then equation* (3) *holds. For every integer $r \ge 1$ satisfying $\sigma > 1/h$ and $r\sigma > 2$, the regularized determinant identity equation* (4) *holds.*

Proof. Each block has the sole nonzero eigenvalue λ_m by lemma 2.2. Since $\sigma > 0$ on every bounded domain, $|\lambda_m| = m^{-\sigma/2}$ is strictly decreasing with m; hence no two blocks contribute the same eigenvalue. Compactness adds only zero to the spectrum.

If $k\sigma > 2$, proposition 3.2 makes the relevant powers trace class, and absolute convergence permits the block traces to be summed:

$$\mathrm{Tr}(T^k) = \sum_{m\in\mathcal{F}_h} \lambda^k_m = \prod_p \sum_{e=0}^{h-1} p^{-eks/2}$$

Table 1: The common cyclic ledger and the distinct metric/angular ledgers.

Quantity	Saturated $S_{h,s}$	Modulo $M_{h,s}$
Bounded domain	$\sigma > 0$	$\sigma > 1/h$
Nonzero eigenvalues	$m^{-s/2}$, $m \in \mathcal{F}_h$, simple	
Block singular square	$m^{-\sigma} \prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1}$	$m^{-\sigma}\zeta(h\sigma)$
Riesz norm	$\prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1/2}$	$\zeta(h\sigma)^{1/2}$
Similarity to normal	$\sigma > 1$	$\sigma > 1/h$

$$= \prod_p \frac{1 - p^{-hks/2}}{1 - p^{-ks/2}} = \frac{\zeta(ks/2)}{\zeta(hks/2)}.$$

At $k\sigma = 2$, the modulus sum is the divergent h-free harmonic series, so the equality line is excluded.

The determinant statement follows from the common simple nonzero eigenvalues and the regularized determinant product in Simon [Sim05, Ch. 9, pp. 75–80]; the exact product and its legality are recorded in appendix B. Nothing in that argument implies similarity. □

4 Isospectrality without bounded similarity

The spectral idempotent for the nonzero eigenvalue in block m is

$$\Pi_{T,m} = \lambda_m^{-1} T_m. \tag{16}$$

It is an oblique projection onto $\mathbb{C}e_m$, and therefore

$$\|\Pi_{T,m}\| = \frac{\rho_T(m)}{|\lambda_m|}. \tag{17}$$

Combining this ratio with the two block masses yields

$$\|\Pi_{S,m}\| = \prod_{p \in J_h(m)} (1 - p^{-\sigma})^{-1/2}, \tag{18}$$

$$\|\Pi_{M,m}\| = \sqrt{\zeta(h\sigma)}. \tag{19}$$

The following block lemma supplies the converse that a mere appeal to necessary projection bounds would miss.

Lemma 4.1 (Uniform block diagonalization). *Let $T = \bigoplus_m T_m$ be a compact orthogonal direct sum of rank-one blocks. Suppose each block has nonzero eigenvalue λ_m with range $\mathbb{C}e_m$, and suppose that the λ_m are pairwise distinct. Then T is boundedly similar to the normal diagonal operator with entries λ_m and zeros on the block kernels if and only if $\sup_m \|\lambda_m^{-1} T_m\| < \infty$.*

Proof. If $T = XNX^{-1}$ and N is normal, the nonzero spectral projections of N are orthogonal. Hence those of T have norms at most $\|X\|\|X^{-1}\|$.

Conversely, decompose a block as $\mathcal{H}_m = \mathbb{C}e_m \oplus K_m$. It has the matrix form

$$T_m = \begin{pmatrix} \lambda_m & \varphi_m \\ 0 & 0 \end{pmatrix}.$$

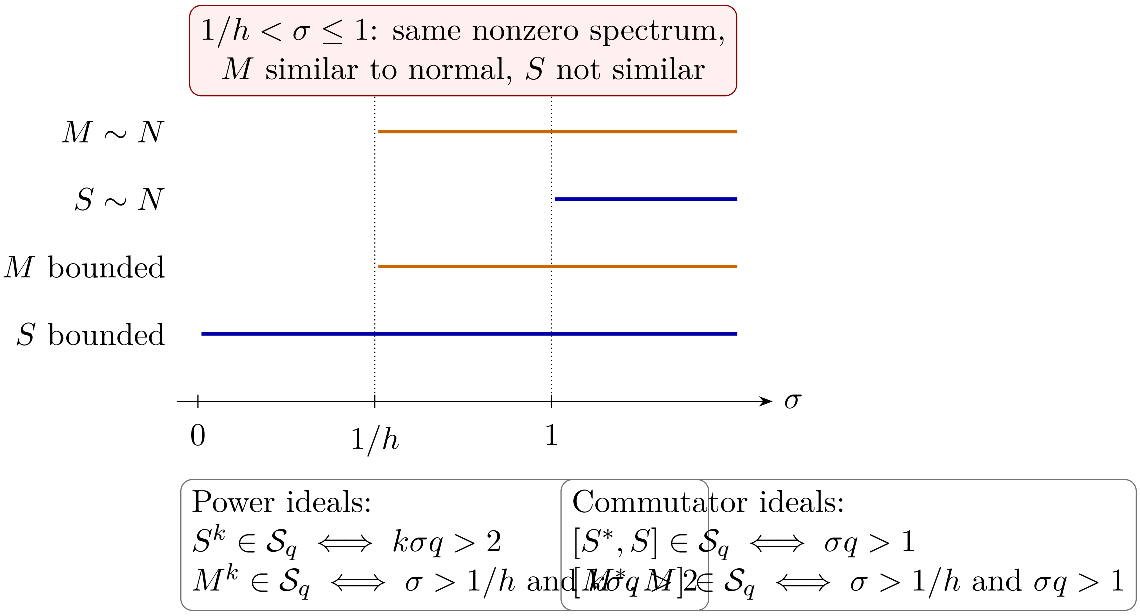

Figure 2: Exact existence, similarity, power-ideal, and commutator-ideal conditions. The horizontal bands encode only the fixed ordering $0 < 1/h \le 1$; the ideal thresholds are stated algebraically because their relative positions depend on h, k, q. Every displayed wall is strict.

Set

$$Y_m = \begin{pmatrix} 1 & -\lambda_m^{-1}\varphi_m \\ 0 & I_{K_m} \end{pmatrix}.$$

A direct multiplication gives $Y_m^{-1} T_m Y_m = \operatorname{diag}(\lambda_m, 0)$. The norm of $\lambda_m^{-1}\varphi_m$ is controlled by the norm of the idempotent $\lambda_m^{-1}T_m$; therefore the hypothesis uniformly bounds both Y_m and Y_m^{-1}. The direct sum $Y = \bigoplus_m Y_m$ is bounded and invertible and conjugates T to the stated diagonal. Since $|\lambda_m| \to 0$, that diagonal is compact and normal. A norm identity and explicit bounds are given in appendix A. □

Theorem 4.2 (Exact similarity gap)**.**

$$S_{h,s} \sim_{\mathrm{bd}} \textit{compact normal} \iff \sigma > 1,$$
$$M_{h,s} \sim_{\mathrm{bd}} \textit{compact normal} \iff \sigma > 1/h.$$

Proof. The modulo norm in equation (19) is a finite constant, independent of m, throughout the bounded domain. The lemma applies.

Every finite set of primes can occur as $J_h(m)$, by taking $m = \prod_{p \in E} p^{h-1}$. Consequently

$$\sup_{m \in \mathcal{F}_h} \|\Pi_{S,m}\| = \left(\prod_p (1 - p^{-\sigma})^{-1} \right)^{1/2}$$

as a finite value or $+\infty$. This is finite exactly for $\sigma > 1$. The lemma again gives both directions. □

In the full interval $1/h < \sigma \le 1$, both operators exist and have the same simple nonzero spectrum. Their common power traces and determinants also agree whenever the corresponding ideal condition is met; here "determinants" means order-r regularized determinants for integers r satisfying $r\sigma > 2$, never the order-one Fredholm determinant. Yet one operator has uniformly controlled eigenprojections and the other does not. This is an obstruction to bounded similarity, not merely to unitary equivalence.

5 Primorial growth of saturated spectral projections

The failure of uniform similarity for $0 < \sigma \le 1$ has a precise maximal order. Let $p_1 < p_2 < \cdots$ be the primes and $P_k = \prod_{j \le k} p_j$, with $P_0 = 1$. For $x \ge 1$, let $k(x)$ be the largest integer for which

$$P_{k(x)}^{h-1} \le x, \qquad m_x = P_{k(x)}^{h-1}. \tag{20}$$

Proposition 5.1 (Exact optimizer)**.** *For every $x \ge 1$ and $\sigma > 0$,*

$$\max_{\substack{m \le x \\ m \in \mathcal{F}_h}} \|\Pi_{S,m}\| = \prod_{p \mid P_{k(x)}} (1 - p^{-\sigma})^{-1/2}, \tag{21}$$

and m_x is a maximizing label.

Proof. Suppose $m \le x$ has $r = |J_h(m)|$ saturated primes. Their mandatory contribution to m gives

$$\prod_{p \in J_h(m)} p^{h-1} \le m \le x.$$

If $r \ge k(x) + 1$, the left side is at least $P_{k(x)+1}^{h-1} > x$, a contradiction. For fixed r, replacing a saturated prime by a smaller omitted prime decreases the arithmetic cost and increases its factor $(1 - p^{-\sigma})^{-1/2}$. Thus the best r-prime set is $\{p_1, \ldots, p_r\}$. Adding another admissible saturated prime strictly increases the projection norm. The optimum therefore uses $r = k(x)$ and is attained at m_x. □

The exact optimizer turns the similarity wall into a classical prime-product problem. Write $y = p_{k(x)}$. The prime number theorem gives

$$(h-1)\vartheta(y) \sim \log x, \qquad y \sim \frac{\log x}{h-1}. \tag{22}$$

Theorem 5.2 (Three maximal-order regimes)**.** *As $x \to \infty$,*

(a) *if $\sigma > 1$, then*

$$\max_{m \le x,\, m \in \mathcal{F}_h} \|\Pi_{S,m}\| \longrightarrow \sqrt{\zeta(\sigma)};$$

(b) *if $\sigma = 1$, then*

$$\max_{m \le x,\, m \in \mathcal{F}_h} \|\Pi_{S,m}\| \sim \sqrt{\mathrm{e}^{\gamma} \log\log x};$$

(c) *if $0 < \sigma < 1$, then*

$$\log \max_{m \le x,\, m \in \mathcal{F}_h} \|\Pi_{S,m}\| \sim \frac{(h-1)^{\sigma-1}(\log x)^{1-\sigma}}{2(1-\sigma)\log\log x}.$$

Proof. For $\sigma > 1$, the partial Euler products in equation (21) converge to $\prod_p (1 - p^{-\sigma})^{-1/2} = \sqrt{\zeta(\sigma)}$. At $\sigma = 1$, Mertens' product theorem [MV06, Thm. 2.7] gives

$$\prod_{p \le y} (1 - p^{-1})^{-1/2} \sim \sqrt{\mathrm{e}^{\gamma} \log y},$$

and equation (22) implies $\log y \sim \log\log x$.

For $0 < \sigma < 1$, expanding the logarithm and applying partial summation to the prime number theorem [MV06, Thm. 6.9, p. 179],

$$\log \prod_{p \le y} (1 - p^{-\sigma})^{-1/2} = \frac{1}{2} \sum_{p \le y} -\log(1 - p^{-\sigma})$$

$$\sim \frac{1}{2}\sum_{p\le y} p^{-\sigma} \sim \frac{y^{1-\sigma}}{2(1-\sigma)\log y}.$$

Substituting equation (22) produces the factor $(h-1)^{\sigma-1}$. In particular, neither its reciprocal nor its deletion is compatible with the exact optimizer. □

The proof uses no finite maximization. The finite optimizer rows reported in appendix D are implementation checks of proposition 5.1; they are not evidence for the asymptotic regimes.

6 Singular-value Weyl laws and the crossover

The saturated block mass is not a constant multiple of $m^{-\sigma}$, so ordinary h-free counting is insufficient. Define the positive generalized weight

$$w_{h,\sigma}(m) = m \prod_{p\in J_h(m)} (1-p^{-\sigma})^{1/\sigma}. \tag{23}$$

Then equation (11) is exactly $\rho_S(m) = w_{h,\sigma}(m)^{-\sigma/2}$.

6.1 The saturated count

Consider the generalized Dirichlet series

$$F_{h,\sigma}(z) = \sum_{m\in\mathcal{F}_h} w_{h,\sigma}(m)^{-z} = \prod_p L_p(z), \tag{24}$$

where initially $\Re z > 1$ and

$$L_p(z) = \sum_{e=0}^{h-2} p^{-ez} + p^{-(h-1)z}(1-p^{-\sigma})^{-z/\sigma}. \tag{25}$$

Factor the single zeta pole:

$$F_{h,\sigma}(z) = \zeta(z)G_{h,\sigma}(z), \qquad G_{h,\sigma}(z) = \prod_p (1-p^{-z})L_p(z). \tag{26}$$

Lemma 6.1 (Local cancellation and strip)**.** *$G_{h,\sigma}$ is holomorphic in*

$$\Re z > \theta_{h,\sigma} := \max\left\{\frac{1}{h}, \frac{1-\sigma}{h-1}\right\} < 1. \tag{27}$$

At $z=1$ it has the positive value $C_{h,\sigma}$ in equation (6)*.*

Proof. On a compact subset of a right half-plane, expand the final term of equation (25) uniformly for large p. Multiplication by $1-p^{-z}$ cancels the first-order p^{-z} term, leaving

$$(1-p^{-z})L_p(z) = 1 + O(p^{-h\Re z}) + O(p^{-(h-1)\Re z-\sigma}). \tag{28}$$

Both error sums converge locally uniformly precisely when $h\Re z > 1$ and $(h-1)\Re z + \sigma > 1$. The Weierstrass theorem for Euler products gives holomorphy in equation (27). Because $h \ge 2$ and $\sigma > 0$, the boundary lies strictly to the left of one. Substituting $z=1$ gives equation (6); every local factor is positive. □

Lemma 6.2 (Wiener–Ikehara hypotheses in Mellin–Stieltjes form)**.** *Let*

$$\mu_{h,\sigma} = \sum_{m\in\mathcal{F}_h} \delta_{w_{h,\sigma}(m)}, \qquad A_S(x) = \mu_{h,\sigma}((0,x]). \tag{29}$$

This positive measure is locally finite and has only finitely many atoms below one. If μ_+ is its restriction to $[1,\infty)$, then, for $\Re z > 1$,

$$\int_{1^-}^{\infty} x^{-z}\, d\mu_+(x) = F_{h,\sigma}(z) - E_{h,\sigma}(z), \tag{30}$$

where $E_{h,\sigma}$ is a finite entire sum. Moreover,

$$F_{h,\sigma}(z) - E_{h,\sigma}(z) - \frac{C_{h,\sigma}}{z-1} \tag{31}$$

extends holomorphically to a neighborhood of the closed half-plane $\Re z \geq 1$. Thus the classical nondecreasing Mellin–Stieltjes form of Wiener–Ikehara [Kor04, Ch. III, Sec. 4, pp. 124–127] applies to μ_+ with residue $C_{h,\sigma}$.

Proof. The nonsaturated local weights are $p^e > 1$, whereas the saturated local weight $p^{h-1}(1-p^{-\sigma})^{1/\sigma}$ tends to infinity. Hence only finitely many local factors are below any fixed bound. Because every prime has one of the bounded exponents $0,\dots,h-1$, a product of these factors can lie in a fixed compact interval only in finitely many ways. This proves local finiteness and, in particular, finiteness below one.

Removing the atoms below one subtracts the finite entire Dirichlet polynomial $E_{h,\sigma}$, proving equation (30). By equation (26) and lemma 6.1, $F_{h,\sigma} = \zeta G_{h,\sigma}$ throughout a strip crossing $\Re z = 1$, $G_{h,\sigma}(1) = C_{h,\sigma}$, and $\zeta(z) - (z-1)^{-1}$ is holomorphic there. Expanding $G_{h,\sigma}(z) = C_{h,\sigma} + (z-1)H(z)$ near one shows that equation (31) is holomorphic at one and elsewhere in a neighborhood of $\Re z \geq 1$. These are precisely the positivity, convergence, pole, and boundary-regularity hypotheses of the cited theorem. □

Proposition 6.3 (Saturated Weyl law)**.** *For $\sigma > 0$, let*

$$A_S(x) = \#\{m \in \mathcal{F}_h : w_{h,\sigma}(m) \leq x\}.$$

Then

$$A_S(x) \sim C_{h,\sigma} x, \qquad s_n(S_{h,s}) \sim \left(\frac{C_{h,\sigma}}{n}\right)^{\sigma/2}. \tag{32}$$

Proof. The verified hypotheses in lemma 6.2 give $A_S(x) \sim C_{h,\sigma}x$; the finitely many removed atoms contribute only $O(1)$. The number of singular values at least t equals $A_S(t^{-2/\sigma})$. Monotone asymptotic inversion gives the second relation. Normal convergence of the Euler product is detailed in appendix C. □

6.2 Modulo and eigenvalue counts

The elementary h-free density is

$$\#\{m \leq x : m \in \mathcal{F}_h\} \sim \frac{x}{\zeta(h)}. \tag{33}$$

For completeness, the identity $\mathbf{1}_{\mathcal{F}_h}(n) = \sum_{d^h \mid n} \mu(d)$ gives

$$\sum_{n\leq x} \mathbf{1}_{\mathcal{F}_h}(n) = \sum_{d\leq x^{1/h}} \mu(d) \left\lfloor \frac{x}{d^h} \right\rfloor = \frac{x}{\zeta(h)} + O_h(x^{1/h}).$$

Proposition 6.4 (Modulo and eigenvalue Weyl laws)**.** *For $\sigma > 1/h$,*

$$s_n(M_{h,s}) \sim \left(\frac{D_{h,\sigma}}{n}\right)^{\sigma/2}, \qquad D_{h,\sigma} = \frac{\zeta(h\sigma)^{1/\sigma}}{\zeta(h)}, \tag{34}$$

$$|\lambda_n| \sim \left(\frac{1/\zeta(h)}{n}\right)^{\sigma/2}. \tag{35}$$

Proof. By equation (12), $\rho_M(m) \geq t$ exactly when $m \leq \zeta(h\sigma)^{1/\sigma} t^{-2/\sigma}$. Applying equation (33) yields the singular-value count $D_{h,\sigma} t^{-2/\sigma}$. The same count without the fiber multiplier gives the eigenvalue constant $1/\zeta(h)$. Monotone inversion proves equations (34) and (35). $\square$

Corollary 6.5 (Exact crossover)**.** *For every $h \geq 2$, $C_{h,1} = D_{h,1} = 1$.*

Proof. At $\sigma = 1$, the bracket in equation (6) is

$$\sum_{e=0}^{h-2} p^{-e} + p^{-(h-1)}(1-p^{-1})^{-1} = \sum_{e=0}^{\infty} p^{-e} = (1-p^{-1})^{-1}.$$

Every Euler factor in $C_{h,1}$ is therefore one. Independently, $D_{h,1} = \zeta(h)/\zeta(h) = 1$. $\square$

The crossover is an equality, not a sign-change theorem: we make no claim about the ordering of $C_{h,\sigma}$ and $D_{h,\sigma}$ for $\sigma \neq 1$. None of the asymptotic or endpoint statements in this section depends on finite records.

7 Self-commutator ideals

The operator ideal wall arises from the first power of a block singular value. A self-commutator sees its square and hence produces a different threshold.

Lemma 7.1 (Rank-one self-commutator)**.** *Let $T = \rho\, u \otimes v$, where u, v are unit vectors, and let a be the modulus of the unique nonzero eigenvalue. The two possibly zero singular values of $T^*T - TT^*$ are equal to*

$$c = \rho^2\sqrt{1 - a^2/\rho^2}. \tag{36}$$

Proof. $T^*T = \rho^2 v \otimes v$ and $TT^* = \rho^2 u \otimes u$. Their difference is self-adjoint and traceless on $\operatorname{span}\{u, v\}$. Its determinant there is $-\rho^4(1 - |\langle u, v\rangle|^2)$. Since $a = \rho|\langle u, v\rangle|$, the two eigenvalues are $\pm c$, proving the claim. $\square$

For the arithmetic blocks, equations (11) and (12) gives the exact angle ratios

$$\frac{|\lambda_m|^2}{\rho_S(m)^2} = \prod_{p \in J_h(m)} (1 - p^{-\sigma}), \tag{37}$$

$$\frac{|\lambda_m|^2}{\rho_M(m)^2} = \frac{1}{\zeta(h\sigma)}. \tag{38}$$

Theorem 7.2 (Exact commutator wall)**.** *For $0 < q < \infty$,*

$$[S_{h,s}^*, S_{h,s}] \in \mathcal{S}_q \iff \sigma q > 1,$$
$$[M_{h,s}^*, M_{h,s}] \in \mathcal{S}_q \iff \sigma > 1/h \text{ and } \sigma q > 1.$$

Proof. Let c_m be the quantity in equation (36) for block m. For S, $c_m \le \rho_S(m)^2$, so

$$\sum_m c_m^q \le \sum_m \rho_S(m)^{2q} < \infty$$

whenever $\sigma q > 1$, by proposition 3.2 with exponent $2q$.

For necessity when $h \ge 3$, fix a prime p_0 and let $m_r = p_0^{h-1} r$ as $r \ne p_0$ varies over primes. The prime r has exponent one and is not saturated, so $J_h(m_r) = \{p_0\}$. The angle factor in equation (37) is a fixed positive constant and $c_{m_r} \asymp r^{-\sigma}$. The prime sum $\sum_r c_{m_r}^q$ diverges when $\sigma q \le 1$.

That witness is ill typed for $h = 2$, because exponent one is already saturated. Instead fix p_0 and take $m_r = p_0 r$. Now $J_2(m_r) = \{p_0, r\}$, while

$$1 - (1 - p_0^{-\sigma})(1 - r^{-\sigma}) \ge p_0^{-\sigma}.$$

Again $c_{m_r} \asymp r^{-\sigma}$, proving divergence at and below the same wall.

For M, equation (38) makes the angular factor a fixed positive constant throughout the bounded domain. Thus $c_m \asymp m^{-\sigma}$, and the h-free sum $\sum_m c_m^q$ converges exactly for $\sigma q > 1$. The boundedness condition $\sigma > 1/h$ remains mandatory. The factor two from the two singular values per nonzero block does not affect membership. □

The $h = 2$ case also admits a closed Hilbert–Schmidt identity. For a squarefree m, define

$$\Lambda_m = \prod_{p|m} (p^\sigma - 1)^{-1}, \qquad \Delta_m = \prod_{p|m} (1 - p^{-\sigma}).$$

Then $\rho_S(m)^2 = \Lambda_m$ and the two commutator singular values are $\Lambda_m \sqrt{1 - \Delta_m}$.

Proposition 7.3 ($h = 2$ Euler control)**.** *If $\sigma > 1/2$, then*

$$\|[S_{2,s}^*, S_{2,s}]\|_2^2 = 2\left\{\prod_p \left[1 + (p^\sigma - 1)^{-2}\right] - \prod_p \left[1 + \frac{p^{-2\sigma}}{1 - p^{-\sigma}}\right]\right\}. \tag{39}$$

Both products converge separately.

Proof. Summing the two squared singular values gives $2\sum_m \Lambda_m^2(1 - \Delta_m)$. Squarefree multiplicativity yields the first Euler product from $\sum_m \Lambda_m^2$. The local nontrivial factor in $\Lambda_m^2 \Delta_m$ is

$$(p^\sigma - 1)^{-2}(1 - p^{-\sigma}) = \frac{p^{-2\sigma}}{1 - p^{-\sigma}},$$

which gives the second product. Each nonconstant local factor is $O(p^{-2\sigma})$, so both products converge for $\sigma > 1/2$. At equality, theorem 7.2 already gives divergence; no subtraction of divergent Euler products is defined. □

No ideal endpoint in this section is inferred from a finite block. The strict failure comes from an explicit infinite positive subfamily.

8 Independent recomputation, controls, and limitations

The formulas were recomputed along two deliberately disjoint routes. One route enumerated the raw maps, constructed finite fiber matrices, and computed spectra, singular values, powers, Riesz idempotents, and self-commutators directly. The other route began from prime-exponent states and independently derived Euler factors, strict convergence walls, the Tauberian strip and residue, the primorial asymptotics, and the commutator families. A third checker audited the infinite proof certificates, while a finite comparator inspected only the common projected fields. The finite and analytic routes agreed on their declared overlap, and registered hostile mutations were rejected. Exact case counts, input seals, and finite optimizer rows are reproducibility metadata in appendix D, not premises of theorem 1.1.

This separation matters. A growing finite compression cannot establish boundedness at an infinite fiber, a strict Schatten endpoint, a Tauberian continuation, or uniform boundedness of all Riesz projections. Conversely, the analytic proof does not certify that a software implementation parses the two maps correctly. The two forms of evidence answer different questions.

Free-UFD negative control. Let $\mathfrak{M}$ be the free commutative monoid generated by atoms a_j, and assign $N(a_j) = p_j$, the jth rational prime. Define exponent saturation, reduction modulo h, and the weights through this norm. The relabeling $a_j \leftrightarrow p_j$ reproduces every exponent combinatoric and Euler product used above, even though the atoms carry no addition and no semantics as rational primes. Thus the paired theorem detects normed unique factorization and its two retractions; by itself it does not establish rational-prime selectivity. This control receives no contribution credit.

Scope. The results concern the frozen maps equation (1), the coefficient $n^{-s/2}$, and the counting measure on $\mathbb{N}$. We do not claim a theorem for arbitrary weighted composition operators, changed base measures, other coefficient weights, or non-free factorization systems. Equal regularized determinants are used only to expose cyclic blindness to block geometry. They are not interpreted as a self-adjoint spectral determinant or as a Hilbert–Pólya construction. The equality $C_{h,1} = D_{h,1} = 1$ likewise gives no universal ordering away from the crossover.

Conclusion. The two retractions preserve exactly the same h-free eigenlines but store discarded prime exponents in incompatible fibers. That difference shifts the existence wall from 0 to $1/h$, the saturated similarity wall to 1, and the commutator wall to $1/q$, while leaving the common cyclic ledger unchanged wherever it is legal. The primorial and Weyl laws make the gap quantitative: isospectrality here is not an approximation to similarity, but a precise separation between arithmetic fixed points and the geometry of their fibers.

A The block graph transform

We record the elementary similarity lemma with explicit constants. Let K be a Hilbert space and

$$T = \begin{pmatrix} \lambda & \varphi \\ 0 & 0 \end{pmatrix} \quad \text{on } \mathbb{C} \oplus K, \qquad \lambda \neq 0,$$

where $\varphi : K \to \mathbb{C}$. Its spectral idempotent at λ is

$$P = \lambda^{-1} T = \begin{pmatrix} 1 & g \\ 0 & 0 \end{pmatrix}, \qquad g = \lambda^{-1}\varphi.$$

The row-operator norm gives

$$\|P\| = (1 + \|g\|^2)^{1/2}. \tag{40}$$

Define

$$Y = \begin{pmatrix} 1 & -g \\ 0 & I \end{pmatrix}, \qquad Y^{-1} = \begin{pmatrix} 1 & g \\ 0 & I \end{pmatrix}.$$

Then

$$Y^{-1}TY = \begin{pmatrix} \lambda & 0 \\ 0 & 0 \end{pmatrix}, \qquad \|Y\|, \|Y^{-1}\| \le 1 + \|g\|. \tag{41}$$

Thus a uniform bound on the block idempotents is precisely what is needed for the direct sum of the graph transforms to be bounded and invertible.

Conversely, if $T = XNX^{-1}$ and N is compact normal, the Riesz projections Q_m of N at distinct nonzero eigenvalues are orthogonal. The corresponding projections of T are XQ_mX^{-1}, so

$$\sup_m \|XQ_mX^{-1}\| \le \|X\|\|X^{-1}\|.$$

This proves the necessity and sufficiency used in lemma 4.1 without appealing to a finite-dimensional condition number that may depend on the block.

B Legality of the common regularized determinant

For an integer $r \ge 1$ and $T \in \mathcal{S}_r$, the order-r regularized determinant is standard [Sim05, Ch. 9, pp. 75–80]. In terms of the algebraic nonzero eigenvalues, counted with multiplicity,

$$\det_r(I - zT) = \prod_{\lambda \in \mathrm{spec}(T)\setminus\{0\}} \left[(1 - z\lambda) \exp\left(\sum_{j=1}^{r-1} \frac{(z\lambda)^j}{j} \right) \right]. \tag{42}$$

The empty sum at $r = 1$ gives the Fredholm determinant. Membership in $\mathcal{S}_r$ guarantees convergence and an entire function of z.

For the operators in this paper, proposition 3.2 shows that both $S_{h,s}$ and $M_{h,s}$ lie in $\mathcal{S}_r$ exactly on the common legal domain

$$\sigma > 1/h, \qquad r\sigma > 2. \tag{43}$$

Their nonzero eigenvalues are the same simple sequence $\lambda_m = m^{-s/2}$, $m \in \mathcal{F}_h$. Substitution in equation (42) therefore proves

$$\det_r(I - zS_{h,s}) = \det_r(I - zM_{h,s}) \tag{44}$$

on equation (43). When $r = 1$, this condition reads $\sigma > 2$. No product is asserted for either unbounded algebraic map, and determinant equality is not used to infer bounded similarity. In particular, throughout the isospectral nonsimilarity band $1/h < \sigma \le 1$, equality means equality in each common legal integer order $r > 2/\sigma$; order $r = 1$ does not occur there.

C Tauberian boundary details

We make explicit the analytic inputs behind proposition 6.3. The local factor in equation (25) is entire in z for each fixed p, because the bases are positive and powers use the real logarithm. On a compact set $K \subset \{\Re z > \theta_{h,\sigma}\}$, choose $\varepsilon > 0$ so that

$$h\Re z \ge 1 + \varepsilon, \qquad (h-1)\Re z + \sigma \ge 1 + \varepsilon \quad (z \in K).$$

The expansion equation (28) is uniform on K, and the sum of its two majorants is bounded by a constant times $\sum_p p^{-1-\varepsilon}$. Hence the Euler product for $G_{h,\sigma}$ converges normally on K. This proves local uniform convergence and holomorphy, not merely pointwise convergence of formal factors.

At $z = 1$, all local factors are positive and their deviations from one are summable. Thus $G_{h,\sigma}(1) = C_{h,\sigma}$ is finite and strictly positive. Since

$$F_{h,\sigma}(z) = \zeta(z)G_{h,\sigma}(z),$$

$F_{h,\sigma}$ has a simple pole at one with residue $C_{h,\sigma}$ and no other singularity on $\Re z = 1$. Subtracting the polar part gives a holomorphic boundary remainder. The counting measure, finite exceptional atoms, Mellin–Stieltjes transform, and the exact cited theorem are assembled in lemma 6.2; the present appendix supplies only the normal-convergence detail used by that lemma.

Finally, if a decreasing positive sequence has counting function $N(t) \sim Ct^{-\alpha}$ as $t \downarrow 0$, elementary monotone inversion gives its nth term as $(C/n)^{1/\alpha}(1 + o(1))$. Here $\alpha = 2/\sigma$, which produces equation (32). No numerical coefficient fit is involved.

D Compact canonical recomputation record

This appendix retains only the finite rows useful for inspecting the implementation of the exact optimizer. The machine-readable source seals, route inventories, 21 finite records, 15 independently audited infinite certificates, seven-case finite comparison, and hostile-suite transaction metadata are kept in the companion evidence ledger. They are not premises of any theorem in this paper.

Table 2: Exact finite optimizer rows generated from the sealed projection. The supercritical finite row has several ties; the canonical maximizer and the primorial label are both 900.

σ	cutoff x	maximizer	primorial label	ties
2/3	100	36	36	36
1	1000	900	900	900
4/3	10000	900	900	6300, 900, 9900

These three rows test the finite optimizer implementation. They do not establish theorem 5.2, which follows from proposition 5.1 and classical prime asymptotics.

References

[AM23] A. V. Abanin and R. S. Mannanikov. Weighted composition operators on quasi-Banach weighted sequence spaces. *Vladikavkazskii Matematicheskii Zhurnal*, 25(4):5–19, 2023.

[Car90] James W. Carlson. The spectra and commutants of some weighted composition operators. *Transactions of the American Mathematical Society*, 317(2):631–654, 1990.

[dWvdW99] B. M. M. de Weger and C. E. van de Woestijne. On the power-free parts of consecutive integers. *Acta Arithmetica*, 90(4):387–395, 1999.

[Kor04] Jacob Korevaar. *Tauberian Theory: A Century of Developments*, volume 329 of *Grundlehren der mathematischen Wissenschaften*. Springer, Berlin, 2004.

[LK15] Doan Minh Luan and Le Hai Khoi. Weighted composition operators on weighted sequence spaces. In *Function Spaces in Analysis*, volume 645 of *Contemporary*

Mathematics, pages 199–215. American Mathematical Society, Providence, RI, 2015.

[MV06] Hugh L. Montgomery and Robert C. Vaughan. *Multiplicative Number Theory I: Classical Theory*, volume 97 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2006.

[Sim05] Barry Simon. *Trace Ideals and Their Applications*, volume 120 of *Mathematical Surveys and Monographs*. American Mathematical Society, Providence, RI, 2 edition, 2005.